

CELL BIOLOGY - DISEASE & CLINICAL VIGNETTE SUMMARY

Revised source-only sheet: presentation/findings are included whenever the slides or textbook actually state them.

Important: A blank or limited presentation column means the uploaded sources did not give a fuller symptom list. This version does **not** invent classic findings from outside knowledge. Slide-image labels are treated as source material.

1. ER & Golgi Disorders

Condition	Source-listed presentation / findings	Organelle / pathway	Defect / mechanism	High-yield recognition
Jaundice	Yellowish discoloration of the skin. Newborn jaundice may occur when hepatic SER is underdeveloped and bilirubin cannot be converted readily for excretion.	Smooth ER; hepatocyte bilirubin handling	Bilirubin accumulates; normally metabolized by SER enzymes in liver cells and excreted in bile.	Yellow skin + bilirubin + SER
Osteogenesis imperfecta	Slide text: very weak bone tissue. Slide image additionally labels blue sclera, multiple fractures, hearing loss, and opalescent teeth.	Rough ER / ERAD protein quality control	Bone cells synthesize and secrete defective procollagen that cannot assemble properly.	OI findings + defective procollagen + RER/ERAD
I-cell disease	No specific bedside symptom list is supplied. Source table identifies skeleton and nervous system as major affected systems.	Golgi lysosomal-enzyme targeting	AR deficiency of N-acetylglucosamine-phosphotransferase -> defective mannose-6-phosphate tagging.	Golgi + M6P targeting defect

2. Lysosomal Storage Disorders

Condition	Source-listed presentation / findings	Organelle / pathway	Defect / mechanism	High-yield recognition
Hurler syndrome (MPS I)	No symptom list supplied; source table lists skeleton and nervous system as major affected systems.	Lysosomal digestion	Faulty enzyme: alpha-L-iduronidase.	MPS I + alpha-L-iduronidase + skeleton/nervous system
McArdle syndrome*	No symptom list supplied; source table lists skeletal muscle as the main tissue affected.	Listed in source lysosomal-storage table	Faulty enzyme: muscle phosphorylase.	Muscle phosphorylase + skeletal muscle
Tay-Sachs disease	No symptom list supplied; source table identifies the nervous system as the major affected system.	Lysosomal digestion	Faulty enzyme listed as GM2-gangliosidase in the source table.	GM2 + nervous system
Gaucher disease	No symptom list supplied; source table identifies liver and spleen as major organs affected.	Lysosomal digestion	Faulty enzyme: glucocerebrosidase.	Glucocerebrosidase + liver/spleen
General lysosomal storage mechanism	Cells may accumulate large secondary lysosomes or residual bodies filled with indigestible macromolecule; this may interfere with tissue function.	Lysosomes	Missing/defective digestive enzyme or faulty posttranslational processing -> substrate accumulation.	Enzyme defect -> stored substrate

*McArdle syndrome is kept in this section because the uploaded textbook table places it there; this sheet preserves source framing.

3. Proteasome, Mitochondrial & Peroxisomal Disorders

Condition	Source-listed presentation / findings	Organelle / pathway	Defect / mechanism	High-yield recognition
Alzheimer disease	Source gives neurodegeneration from damaging protein aggregates; no more specific clinical symptom list is supplied.	Proteasome / protein quality control	Failure of proteasomes or related quality control permits protein aggregates that can damage or kill cells.	Protein aggregates + brain neurodegeneration
Huntington disease	Textbook vignette: progressive uncoordination; depression; apathy; dysarthria; saccadic extraocular eye movements; hyperactive gag reflex; increased tone. Imaging: striatal/caudate atrophy with "boxcar ventricles" . MMSE 24/30; CAG repeats 20/49.	Protein aggregates; mitochondrial bioenergetics also tested	Protein aggregates are emphasized; assessment links disease to mitochondrial membrane potential/bioenergetic failure.	Movement + psychiatric/cognitive findings + caudate/striatal atrophy + CAG

Condition	Source-listed presentation / findings	Organelle / pathway	Defect / mechanism	High-yield recognition
MERRF	Name itself supplies myoclonic epilepsy with ragged-red fibers ; source also notes structural abnormalities in skeletal muscle and other cells.	Mitochondria / mtDNA	Mutated mitochondrial lysine-tRNA -> defective synthesis of respiratory-chain proteins; maternal mtDNA context.	Myoclonic epilepsy + ragged-red fibers + mtDNA tRNA
Neonatal adrenoleukodystrophy	Severe neurologic symptoms are stated; VLCFA accumulation can disrupt myelin sheaths .	Peroxisome / VLCFA transport and beta-oxidation	Defective integral membrane protein needed for VLCFA transport into peroxisome.	VLCFA + myelin + severe neurologic symptoms
Zellweger syndrome	Source says it affects the structure and function of several organ systems ; no specific symptom list is supplied.	Peroxisomes	Deficiencies of peroxisomal enzymes / proteins; peroxisome deficiency also causes failed VLCFA oxidation and bile-acid precursor accumulation.	Peroxisome deficiency + multisystem disease

4. Cytoskeletal, Receptor & Inclusion-Related Conditions

Condition	Source-listed presentation / findings	Organelle / pathway	Defect / mechanism	High-yield recognition
Chediak-Higashi syndrome	Described as an autosomal recessive immunodeficiency disorder . No more detailed symptom list is supplied in the slide.	Microtubules / intracellular transport	Defective microtubule polymerization -> abnormal intracellular protein transport -> delayed phagosome-lysosome fusion in leukocytes.	Immunodeficiency + microtubules + delayed phagosome-lysosome fusion
Epidermolysis bullosa simplex	Textbook vignette: skin blisters easily with rubbing/scratching ; blisters mainly on hands and feet and heal without scars .	Intermediate filaments / keratin	KRT5 and KRT14 mutations; loss of keratin mechanical support.	Friction blisters + hands/feet + no scars + keratin 5/14
Pseudohypoparathyroidism	Source presentation is functional rather than symptomatic: parathyroid hormone is produced, but target cells cannot respond .	Plasma-membrane receptor signaling	Nonfunctioning parathyroid-hormone receptor.	Hormone present + receptor failure
One type of dwarfism	The source identifies dwarfism and states growth hormone is produced but target cells cannot respond.	Plasma-membrane receptor signaling	Nonfunctioning growth-hormone receptor.	Growth hormone present + receptor failure
Hemosiderosis	Widespread cellular hemosiderin; source states hemosiderosis itself does not damage cell or organ function .	Cytoplasmic inclusion / iron storage	May follow increased dietary iron, impaired iron utilization, or excessive RBC lysis.	Widespread hemosiderin without intrinsic tissue damage
Hemochromatosis / iron overload syndrome	Liver and other tissues can be damaged when iron accumulation becomes extreme.	Cytoplasmic inclusion / iron overload	Extreme accumulation of iron in cellular hemosiderin.	Extreme iron + organ injury

5. Cancer-Related Clinical Correlations

Condition	Source-listed presentation / findings	Organelle / pathway	Defect / mechanism	High-yield recognition
High-risk HPV-associated cervical cancer	Slide vignette: 42-year-old woman ; two Pap smears show dysplastic epithelial cells ; molecular testing detects high-risk HPV E6/E7 DNA.	Apoptosis / p53	E6 binds p53 and causes degradation; slide question identifies prevention of apoptosis as the cancer-promoting mechanism.	Dysplastic Pap + high-risk HPV + E6 -> p53 loss
Astrocytoma	No clinical symptom vignette supplied. Textbook gives a diagnostic marker application .	Intermediate filaments	GFAP immunocytochemistry can identify astrocytic tumor origin.	GFAP -> astrocytoma/astrocytic origin

6. Symptom / Finding -> Disease Pattern Recognition

Source-listed clue	Think of...
Yellow skin / bilirubin accumulation	Jaundice
Blue sclera + multiple fractures + hearing loss + opalescent teeth	Osteogenesis imperfecta
Uncoordination + depression/apathy + dysarthria + saccadic eye movements + increased tone + caudate/striatal atrophy	Huntington disease
Myoclonic epilepsy + ragged-red fibers	MERRF

Source-listed clue	Think of...
Severe neurologic symptoms + VLCFA accumulation + disrupted myelin	Neonatal adrenoleukodystrophy
Immunodeficiency + delayed phagosome-lysosome fusion	Chediak-Higashi syndrome
Friction-induced blisters on hands/feet that heal without scars	Epidermolysis bullosa simplex
Dysplastic Pap smears + high-risk HPV E6/E7 + p53 degradation	HPV-associated cervical cancer
Widespread hemosiderin without direct tissue damage	Hemosiderosis
Extreme hemosiderin iron accumulation + liver/other-organ injury	Hemochromatosis / iron overload

Source boundary: This revised sheet only includes manifestations that were explicitly stated in the uploaded slides/textbook or visibly labeled on a slide image. Classic findings not present in the supplied material are intentionally omitted.